Week 9: Sales and Customer Analytics System

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The phone case e-commerce company sells different types of mobile phone cases to customers through its online platform. The company maintains customer, product, order, payment, and sales information in its database. SQL aggregate functions and analytical queries are used to calculate total sales, average order value, and customer spending. These queries help identify top customers, best-selling phone cases, and profitable product categories. The reports help the management understand sales performance, customer purchasing behaviour, and make better business decisions to improve the company's sales and profitability.

1. Apply SQL Aggregate Functions

Aggregate functions perform calculations on multiple records and return a single summarized value.

Functions Covered: Function	Purpose	Example
COUNT()	Counts number of records	Total customers/orders
SUM()	Calculates total value	Total sales amount
AVG()	Finds average value	Average order value
MIN()	Finds minimum value	Lowest product price
MAX()	Finds maximum value	Highest product

-
```
TOTAL_ORDERS
------------
          17
```

-
```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_REVENUE
  2  FROM ORDERS;

TOTAL_REVENUE
-------------
        11670
```

-
```
SQL> SELECT AVG(TOTAL_AMOUNT) AS AVERAGE_ORDER_VALUE
  2  FROM ORDERS;

AVERAGE_ORDER_VALUE
-------------------
         686.470588
```

-
```
SQL> SELECT MAX(TOTAL_AMOUNT) AS HIGHEST_ORDER_AMOUNT,
  2         MIN(TOTAL_AMOUNT) AS LOWEST_ORDER_AMOUNT
  3  FROM ORDERS;

HIGHEST_ORDER_AMOUNT LOWEST_ORDER_AMOUNT
-------------------- -------------------
                1599                 199
```

-
```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SALES
  2  FROM ORDERS
  3  WHERE ORDER_DATE >= DATE '2026-09-01'
  4  AND ORDER_DATE < DATE '2026-10-01';

TOTAL_SALES
-----------
      11421
```

2. Generate Total Sales Reports

1. Total number of orders placed

```
TOTAL_ORDERS
------------
          17
```

2. Total revenue generated from sales

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_REVENUE
  2  FROM ORDERS;

TOTAL_REVENUE
-------------
        11670
```

3. Average order value

```
SQL> SELECT AVG(TOTAL_AMOUNT) AS AVERAGE_ORDER_VALUE
  2  FROM ORDERS;

AVERAGE_ORDER_VALUE
-------------------
         686.470588
```

4. Highest and lowest order amount

```
SQL> SELECT MAX(TOTAL_AMOUNT) AS HIGHEST_ORDER_AMOUNT,
  2         MIN(TOTAL_AMOUNT) AS LOWEST_ORDER_AMOUNT
  3  FROM ORDERS;

HIGHEST_ORDER_AMOUNT LOWEST_ORDER_AMOUNT
-------------------- -------------------
                1599                 199
```

5. Total sales for a specific period

Example: September 20 to September 30, 2026

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SALES
  2  FROM ORDERS
  3  WHERE ORDER_DATE >= DATE '2026-09-01'
  4  AND ORDER_DATE < DATE '2026-10-01';

TOTAL_SALES
-----------
      11421
```

3. Customer Purchase Analysis

1. Total number of orders placed by each customer

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         COUNT(O.ORDER_ID) AS TOTAL_ORDERS
  4  FROM CUSTOMER C
  5  LEFT JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  ORDER BY TOTAL_ORDERS DESC;

CUSTO CUSTOMER_NAME                  TOTAL_ORDERS
----- ------------------------------ ------------
C005  Karthik S                                 4
C004  Divya Kumar                               4
C002  Aakash                                    3
C001  Arun Kumar                                3
C003  Rahul Raj                                 3
C006  Priya Sharma                              0

6 rows selected.
```

2. Total amount spent by each customer

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         NVL(SUM(O.TOTAL_AMOUNT), 0) AS AMOUNT_SPENT
  4  FROM CUSTOMER C
  5  LEFT JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  ORDER BY AMOUNT_SPENT DESC;

CUSTO CUSTOMER_NAME                  AMOUNT_SPENT
----- ------------------------------ ------------
C002  Aakash                               2598
C005  Karthik S                            2548
C001  Arun Kumar                           2447
C003  Rahul Raj                            2348
C004  Divya Kumar                          1729
C006  Priya Sharma                            0

6 rows selected.
```

3. Average spending per customer

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         NVL(AVG(O.TOTAL_AMOUNT), 0) AS AVERAGE_SPENDING
  4  FROM CUSTOMER C
  5  LEFT JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  ORDER BY AVERAGE_SPENDING DESC;

CUSTO CUSTOMER_NAME                  AVERAGE_SPENDING
----- ------------------------------ ----------------
C002  Aakash                                      866
C001  Arun Kumar                          815.666667
C003  Rahul Raj                           782.666667
C005  Karthik S                                  637
C004  Divya Kumar                             432.25
C006  Priya Sharma                                 0

6 rows selected.
```

4. Customers with maximum purchase amount

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  HAVING SUM(O.TOTAL_AMOUNT) = (
  9      SELECT MAX(TOTAL_SPENT)
 10      FROM (
 11          SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SPENT
 12          FROM ORDERS
 13          GROUP BY CUSTOMER_ID
 14      )
 15  );

CUSTO CUSTOMER_NAME                 TOTAL_SPENDING
----- ----------------------------- --------------
C002  Aakash                                2598
```

5. Customers with fewer purchases

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         COUNT(O.ORDER_ID) AS TOTAL_ORDERS
  4  FROM CUSTOMER C
  5  LEFT JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  HAVING COUNT(O.ORDER_ID) < 2
  9  ORDER BY TOTAL_ORDERS ASC;

CUSTO CUSTOMER_NAME                  TOTAL_ORDERS
----- ----------------------------- ------------
C006  Priya Sharma                              0
```

4. Find Top Customers Based on Purchase Amount

1. Top 5 customers based on total spending

```
SQL> SELECT CUSTOMER_ID,
  2         CUSTOMER_NAME,
  3         TOTAL_SPENDING
  4  FROM (
  5      SELECT C.CUSTOMER_ID,
  6             C.CUSTOMER_NAME,
  7             SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING
  8      FROM CUSTOMER C
  9      JOIN ORDERS O
 10      ON C.CUSTOMER_ID = O.CUSTOMER_ID
 11      GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
 12      ORDER BY TOTAL_SPENDING DESC
 13  )
 14  WHERE ROWNUM <= 5;

CUSTO CUSTOMER_NAME                 TOTAL_SPENDING
----- ----------------------------- --------------
C002  Aakash                                2598
C005  Karthik S                             2548
C001  Arun Kumar                            2447
C003  Rahul Raj                             2348
C004  Divya Kumar                           1729
```

2. Customers with maximum number of orders

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         COUNT(O.ORDER_ID) AS TOTAL_ORDERS
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  HAVING COUNT(O.ORDER_ID) = (
  9      SELECT MAX(ORDER_COUNT)
 10      FROM (
 11          SELECT COUNT(*) AS ORDER_COUNT
 12          FROM ORDERS
 13          GROUP BY CUSTOMER_ID
 14      )
 15  );

CUSTO CUSTOMER_NAME                 TOTAL_ORDERS
----- ----------------------------- ------------
C005  Karthik S                                4
C004  Divya Kumar                              4
```

3. Frequent customers

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         COUNT(O.ORDER_ID) AS TOTAL_ORDERS
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  HAVING COUNT(O.ORDER_ID) >= 3
  9  ORDER BY TOTAL_ORDERS DESC;

CUSTO CUSTOMER_NAME                 TOTAL_ORDERS
----- ----------------------------- ------------
C005  Karthik S                                4
C004  Divya Kumar                              4
C001  Arun Kumar                               3
C002  Aakash                                   3
C003  Rahul Raj                                3
```

4. High-value customers

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         COUNT(O.ORDER_ID) AS TOTAL_ORDERS
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  HAVING COUNT(O.ORDER_ID) > (
  9      SELECT AVG(ORDER_COUNT)
 10      FROM (
 11          SELECT COUNT(ORDER_ID) AS ORDER_COUNT
 12          FROM ORDERS
 13          GROUP BY CUSTOMER_ID
 14      )
 15  )
 16  ORDER BY TOTAL_ORDERS DESC;

CUSTO CUSTOMER_NAME                 TOTAL_ORDERS
----- ----------------------------- ------------
C004  Divya Kumar                              4
C005  Karthik S                                4
```

5. Identify Best-Selling Products

1. Products with maximum sales quantity

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         P.PRODUCT_ID,
  4         P.PRODUCT_NAME,
  5         SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD
  6  FROM CUSTOMER C
  7  JOIN ORDERS O
  8  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  9  JOIN ORDER_DETAILS OD
 10  ON O.ORDER_ID = OD.ORDER_ID
 11  JOIN PRODUCT P
 12  ON OD.PRODUCT_ID = P.PRODUCT_ID
 13  GROUP BY C.CUSTOMER_ID,
 14           C.CUSTOMER_NAME,
 15           P.PRODUCT_ID,
 16           P.PRODUCT_NAME
 17  ORDER BY TOTAL_QUANTITY_SOLD DESC;

CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
C003  Rahul Raj                      P000004
Floral Printed Case
                  4

C005  Karthik S                      P000002
Soft Silicone Case
                  2

CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------

C001  Arun Kumar                     P000001
Clear Shock Case
                  2

C002  Aakash                         P000003
Crystal Clear Case

CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
                  2

C005  Karthik S                      P000010
Slim Basic Case
                  1

C001  Arun Kumar                     P000005
```

```
CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
C005  Karthik S                      P000024
Collage Photo Case
                  1

C003  Rahul Raj                      P000008
Matte Basic Case
                  1

CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------

C004  Divya Kumar                    P000022
Luxury Gold Case
                  1

C004  Divya Kumar                    P000006
Personal Photo Case

CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
                  1

C002  Aakash                         P000007
Heavy Duty Case
                  1

C005  Karthik S                      P000005

CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
Luxury Designer Case
                  1

C004  Divya Kumar                    P000009
Classic Black Case
                  1


CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
C004  Divya Kumar                    P000001
Clear Shock Case
                  1
```

```
C001  Arun Kumar                     P000002
Soft Silicone Case
                  1

CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------


16 rows selected.
```

2. Products generating highest revenue

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         P.PRODUCT_ID,
  4         P.PRODUCT_NAME,
  5         SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  6         SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  7  FROM CUSTOMER C
  8  JOIN ORDERS O
  9  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 10  JOIN ORDER_DETAILS OD
 11  ON O.ORDER_ID = OD.ORDER_ID
 12  JOIN PRODUCT P
 13  ON OD.PRODUCT_ID = P.PRODUCT_ID
 14  GROUP BY C.CUSTOMER_ID,
 15           C.CUSTOMER_NAME,
 16           P.PRODUCT_ID,
 17           P.PRODUCT_NAME
 18  ORDER BY TOTAL_REVENUE DESC;

CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
C003  Rahul Raj                      P000004
Floral Printed Case
            4          3599

C002  Aakash                         P000003
Crystal Clear Case
            2          2099

CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------

C005  Karthik S                      P000005
Luxury Designer Case
            1           999

C001  Arun Kumar                     P000001
Clear Shock Case

CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
            2           998

C005  Karthik S                      P000002
Soft Silicone Case
            2           750

C001  Arun Kumar                     P000005
```

```
CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
Luxury Designer Case
            1           699

C004  Divya Kumar                    P000006
Personal Photo Case
            1           550


CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
C005  Karthik S                      P000024
Collage Photo Case
            1           550

C004  Divya Kumar                    P000022
Luxury Gold Case
            1           500

CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------

C002  Aakash                         P000007
Heavy Duty Case
            1           499

C004  Divya Kumar                    P000001
Clear Shock Case

CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
            1           450

C001  Arun Kumar                     P000006
Personal Photo Case
            1           449

C001  Arun Kumar                     P000002

CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
Soft Silicone Case
            1           301

C005  Karthik S                      P000010
Slim Basic Case
```

```
CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
Soft Silicone Case
            1           301

C005  Karthik S                      P000010
Slim Basic Case
            1           249


CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
C004  Divya Kumar                    P000009
Classic Black Case
            1           229

C003  Rahul Raj                      P000008
Matte Basic Case
            1           199

CUSTO CUSTOMER_NAME                  PRODUCT
----- ------------------------------ -------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------


16 rows selected.
```

3. Least-selling products

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD
  4  FROM PRODUCT P
  5  JOIN ORDER_DETAILS OD
  6  ON P.PRODUCT_ID = OD.PRODUCT_ID
  7  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  8  HAVING SUM(OD.QUANTITY) = (
  9      SELECT MIN(TOTAL_QUANTITY)
 10      FROM (
 11          SELECT SUM(QUANTITY) AS TOTAL_QUANTITY
 12          FROM ORDER_DETAILS
 13          GROUP BY PRODUCT_ID
 14      )
 15  );

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
P000008
Matte Basic Case
                  1

P000010
Slim Basic Case
                  1

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------

P000009
Classic Black Case
                  1

P000007
Heavy Duty Case

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
                  1

P000022
Luxury Gold Case
                  1

P000024

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
Collage Photo Case
                  1


6 rows selected.
```

4. Products requiring promotion

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         P.PRICE,
  4         P.STOCK_QUANTITY
  5  FROM PRODUCT P
  6  WHERE P.STOCK_QUANTITY > 0
  7  AND NOT EXISTS (
  8      SELECT 1
  9      FROM ORDER_DETAILS OD
 10      WHERE OD.PRODUCT_ID = P.PRODUCT_ID
 11  )
 12  ORDER BY P.PRODUCT_NAME;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P000019
Abstract Art Case
       379             25

P000011
Blue Silicone Case
       279             45

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------

P000018
Cartoon Print Case
       329             35

P000014
Clear Glitter Case

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
       249             50

P000015
Crystal Edge Case
       269             45

P000025

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
Custom Memory Case
       479             30
```

```
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
       599             15

P000012
Pink Silicone Case
       279             40

P000021

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
Premium Pattern Case
       649             15

P000027
Rugged Protection Case
       549             20


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
P000026
Shockproof Armor Case
       499             25

P000030
Simple White Case
       239             25

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------

P000013
Soft Grip Case
       299             35

P000016
Ultra Clear Case

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE STOCK_QUANTITY
---------- --------------
       229             55


18 rows selected.
```

6. Perform Category-Wise Sales Analysis

1. Total sales generated by each category

```
SQL> SELECT C.CATEGORY_NAME,
  2         SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_SALES
  3  FROM CATEGORY C
  4  JOIN PRODUCT P
  5  ON C.CATEGORY_ID = P.CATEGORY_ID
  6  JOIN ORDER_DETAILS OD
  7  ON P.PRODUCT_ID = OD.PRODUCT_ID
  8  GROUP BY C.CATEGORY_NAME
  9  ORDER BY TOTAL_SALES DESC;

CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_SALES
-----------
Printed Cases
       3599

Designer Cases
       2198

Basic Cases
       2125


CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_SALES
-----------
Transparent Cases
       2099

Photo Cases
       1549

Silicone Cases
       1051


CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_SALES
-----------
Protective Cases
        499


7 rows selected.
```

2. Number of products sold in each category

```
SQL> SELECT C.CATEGORY_ID,
  2         C.CATEGORY_NAME,
  3         NVL(SUM(OD.QUANTITY), 0) AS TOTAL_PRODUCTS_SOLD
  4  FROM CATEGORY C
  5  LEFT JOIN PRODUCT P
  6  ON C.CATEGORY_ID = P.CATEGORY_ID
  7  LEFT JOIN ORDER_DETAILS OD
  8  ON P.PRODUCT_ID = OD.PRODUCT_ID
  9  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
 10  ORDER BY TOTAL_PRODUCTS_SOLD DESC;

CATEGOR
-------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD
-------------------
CAT0001
Basic Cases
                  6

CAT0004
Printed Cases
                  4

CATEGOR
-------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD
-------------------

CAT0002
Silicone Cases
                  3

CAT0005
Designer Cases

CATEGOR
-------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD
-------------------
                  3

CAT0006
Photo Cases
                  3

CAT0003

CATEGOR
-------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD
-------------------
Transparent Cases
                  2

CAT0007
Protective Cases
                  1


7 rows selected.
```

3. Highest revenue-generating category

```
SQL> SELECT C.CATEGORY_ID,
  2         C.CATEGORY_NAME,
  3         SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  4  FROM CATEGORY C
  5  JOIN PRODUCT P
  6  ON C.CATEGORY_ID = P.CATEGORY_ID
  7  JOIN ORDER_DETAILS OD
  8  ON P.PRODUCT_ID = OD.PRODUCT_ID
  9  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
 10  HAVING SUM(OD.QUANTITY * OD.PRICE) = (
 11      SELECT MAX(CATEGORY_REVENUE)
 12      FROM (
 13          SELECT SUM(OD2.QUANTITY * OD2.PRICE) AS CATEGORY_REVENUE
 14          FROM PRODUCT P2
 15          JOIN ORDER_DETAILS OD2
 16          ON P2.PRODUCT_ID = OD2.PRODUCT_ID
 17          GROUP BY P2.CATEGORY_ID
 18      )
 19  );

CATEGOR
-------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
CAT0004
Printed Cases
         3599
```

4. Average sales per category

```
SQL> SELECT C.CATEGORY_ID,
  2         C.CATEGORY_NAME,
  3         NVL(AVG(OD.QUANTITY * OD.PRICE), 0) AS AVERAGE_SALES
  4  FROM CATEGORY C
  5  LEFT JOIN PRODUCT P
  6  ON C.CATEGORY_ID = P.CATEGORY_ID
  7  LEFT JOIN ORDER_DETAILS OD
  8  ON P.PRODUCT_ID = OD.PRODUCT_ID
  9  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
 10  ORDER BY AVERAGE_SALES DESC;

CATEGOR
------
CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_SALES
-------------
CAT0004
Printed Cases
       1799.5

CAT0003
Transparent Cases
       1049.5

CATEGOR
-------
CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_SALES
-------------

CAT0005
Designer Cases
   732.666667

CAT0002
Silicone Cases

CATEGOR
-------
CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_SALES
-------------
        525.5

CAT0006
Photo Cases
   516.333333

CAT0007

CATEGOR
------
CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_SALES
-------------
Protective Cases
         499

CAT0001
Basic Cases
         425


7 rows selected.
```

7. Generate Business Analytics Reports

Report 1: Sales Performance Report

- **Total sales**
- **Number of orders**
- **Average order value**
- **Highest order value**

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SALES,
  2         COUNT(ORDER_ID) AS NUMBER_OF_ORDERS,
  3         AVG(TOTAL_AMOUNT) AS AVERAGE_ORDER_VALUE,
  4         MAX(TOTAL_AMOUNT) AS HIGHEST_ORDER_VALUE
  5  FROM ORDERS;

TOTAL_SALES NUMBER_OF_ORDERS AVERAGE_ORDER_VALUE HIGHEST_ORDER_VALUE
----------- ---------------- ------------------- -------------------
      11670               17          686.470588                1599
```

Report 2: Customer Analytics Report

- Customer name
- Number of orders
- Total spending
- Purchase frequency

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS,
  4         NVL(SUM(O.TOTAL_AMOUNT), 0) AS TOTAL_SPENDING,
  5         CASE
  6             WHEN COUNT(O.ORDER_ID) >= 3 THEN 'Frequent Customer'
  7             WHEN COUNT(O.ORDER_ID) >= 1 THEN 'Regular Customer'
  8             ELSE 'No Purchases'
  9         END AS PURCHASE_FREQUENCY
 10  FROM CUSTOMER C
 11  LEFT JOIN ORDERS O
 12  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 13  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
 14  ORDER BY TOTAL_SPENDING DESC;

CUSTO CUSTOMER_NAME                  NUMBER_OF_ORDERS TOTAL_SPENDING
----- ------------------------------ ---------------- --------------
PURCHASE_FREQUENC
-----------------
C002  Aakash                                        3           2598
Frequent Customer

C005  Karthik S                                     4           2548
Frequent Customer

C001  Arun Kumar                                    3           2447
Frequent Customer


CUSTO CUSTOMER_NAME                  NUMBER_OF_ORDERS TOTAL_SPENDING
----- ------------------------------ ---------------- --------------
PURCHASE_FREQUENC
-----------------
C003  Rahul Raj                                     3           2348
Frequent Customer

C004  Divya Kumar                                   4           1729
Frequent Customer

C006  Priya Sharma                                  0              0
No Purchases


6 rows selected.
```

Report 3: Product Performance Report

- Product name
- Quantity sold
- Revenue generated

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         NVL(SUM(OD.QUANTITY), 0) AS QUANTITY_SOLD,
  4         NVL(SUM(OD.QUANTITY * OD.PRICE), 0) AS REVENUE_GENERATED
  5  FROM PRODUCT P
  6  LEFT JOIN ORDER_DETAILS OD
  7  ON P.PRODUCT_ID = OD.PRODUCT_ID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  ORDER BY REVENUE_GENERATED DESC;

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
P000004
Floral Printed Case
            4              3599

P000003
Crystal Clear Case
            2              2099

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------

P000005
Luxury Designer Case
            2              1698

P000001
Clear Shock Case

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
            3              1448

P000002
Soft Silicone Case
            3              1051

P000006

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
Personal Photo Case
            2               999

P000024
Collage Photo Case
```

```
PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
P000022
Luxury Gold Case
            1                500

P000007
Heavy Duty Case
            1                499

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------

P000010
Slim Basic Case
            1                249

P000009
Classic Black Case

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
            1                229

P000008
Matte Basic Case
            1                199

P000013

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
Soft Grip Case
            0                  0

P000026
Shockproof Armor Case
            0                  0


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
P000028
```

```
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------ -----------------

P000011
Blue Silicone Case
           0                 0

P000030
Simple White Case

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------ -----------------
           0                 0

P000015
Crystal Edge Case
           0                 0

P000016

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------ -----------------
Ultra Clear Case
           0                 0

P000014
Clear Glitter Case
           0                 0


PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------ -----------------
P000023
Family Photo Case
           0                 0

P000017
Floral Print Case
           0                 0

PRODUCT
-------
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------ -----------------


30 rows selected.
```

Report 4: Category Analysis Report

- Category name
- Total products sold
- Total revenue

```
SQL> SELECT C.CATEGORY_ID,
  2         C.CATEGORY_NAME,
  3         NVL(SUM(OD.QUANTITY), 0) AS TOTAL_PRODUCTS_SOLD,
  4         NVL(SUM(OD.QUANTITY * OD.PRICE), 0) AS TOTAL_REVENUE
  5  FROM CATEGORY C
  6  LEFT JOIN PRODUCT P
  7  ON C.CATEGORY_ID = P.CATEGORY_ID
  8  LEFT JOIN ORDER_DETAILS OD
  9  ON P.PRODUCT_ID = OD.PRODUCT_ID
 10  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
 11  ORDER BY TOTAL_REVENUE DESC;

CATEGOR
-------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------
CAT0004
Printed Cases
                  4          3599

CAT0005
Designer Cases
                  3          2198

CATEGOR
-------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------

CAT0001
Basic Cases
                  6          2125

CAT0003
Transparent Cases

CATEGOR
-------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------
                  2          2099

CAT0006
Photo Cases
                  3          1549

CAT0002

CATEGOR
-------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------
Silicone Cases
                  3          1051

CAT0007
Protective Cases
                  1           499


7 rows selected.
```